

Dummit & Foote 8.1

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1 Problems

The Gaussian integers is a Euclidean domain. Define $N(a + bi) = a^2 + b^2$ as usual. Let $\alpha, \beta \in \mathbb{Z}[i]$, $\beta \neq 0$ and let $\gamma \in \mathbb{Z}[i]$ be closest to α/β . We have $N(\alpha/\beta - \gamma) \leq 1/4 + 1/4 = 1/2$, so $N(\alpha - \beta\gamma) \leq \frac{1}{2}N(\beta) < N(\beta)$, AWD.

Consider when $D = -3$. $N(\alpha/\beta - \gamma) \leq 1/4 + 3/4 = 1$, so $N(\alpha - \beta\gamma) \leq N(\beta)$. Check edge cases?

A discrete valuation ring R is such that there exists a valuation ν on the field of fractions of R such that R is the valuation ring of ν . Ex: R is the set of rationals whose denominators is prime to p . $R^{-1}R$ is $\mathbb{Q}$. ν is the order of a rational (surjective homomorphism, satisfies min condition).

A discrete valuation ring is a Euclidean domain. Let $a, b \in R$, $b \neq 0$. Define N to be ν at nonzero values in K and $N(0) = 0$. If $N(a) < N(b)$, we're done. Else $N(a) \geq N(b)$, so that for $q = ab^{-1}$, $N(q) \geq 0$, or $q \in R$. Write $a = qb$, AWD.

In a Euclidean domain every Ideal is principal. Let $I \triangleleft R$. If $I = 0$, $I = (0)$. Else I has a nonzero element. Let $m \neq 0$ in I have minimal norm. For any $x \in I$, there exists $q, r \in R$ for which $x = qm + r$ and $r = 0$ or $N(r) < N(m)$. Note $r = x - qm \in I$, so by minimality of m we have $r = 0$. That is, $I \subset (m)$. Conversely, $(m) \subset I$, so $I = (m)$.

Ex: Since $(2, x) \subset \mathbb{Z}[x]$ is not principal, $\mathbb{Z}[x]$ is not a Euclidean domain WRT any norm.

Let $R = \mathbb{Z}[\sqrt{-5}]$, $I = (2, 2 + \sqrt{-5})$. Suppose FTSOC $I = (a + b\sqrt{-5})$. $a^2 + 5b^2 \mid 4$ and $a^2 + 5b^2 \mid 9$. Then $a^2 + 5b^2 = 1$, forcing $a = \pm 1$, $b = 0$. This isn't the ideal I .

If a, b are nonzero elements of a commutative ring R such that the ideal generated by a, b is the ideal generated by d , then d is a greatest common divisor of a, b .

Since $(a, b) = (d)$, we have $(a, b) \subset (d)$. Suppose $(a, b) \subset (d')$. Then $a \in (d')$ and $b \in (d')$. Further as $(a, b) = (d)$ there exist $r, s \in R$ such that $ar + bs = d$. Then $ar \in (d')$ and $bs \in (d')$, so $d = ar + bs \in (d')$. It follows $(d) \subset (d')$, AWD.

Prop 3. If $(d) = (d')$, then $d = ud'$ for some $u \in R$ and $d' = vd$ for some $v \in R$. Then $dd' = dd'uv$, so $uv = 1$, so u, v are units. Suppose d and d' are greatest common divisors of $a, b \in R$. We know $(a, b) \subset (d)$ and $(a, b) \subset (d')$. By the maximality condition, $(d) \subset (d')$ and $(d') \subset (d)$, so $(d') = (d)$. The result follows from the previous work.

By induction, $(a, b) = (d)$, so by a previous proposition, d is a greatest common divisor of a, b . By reverse substitution, $d = ax + by$ for some $x, y \in R$.

Rigorous proof: Let our equations be $E_0 : a = bq_0 + r_0$, $E_1 : b = r_0q_1 + r_1$, $E_2 : r_0 = r_1q_2 + r_2$, and so on to $E_n : r_{n-2} = r_{n-1}q_n + r_n$, where $r_n | r_{n-1}$. We claim that for $d = r_n$, $d | a$ and $d | b$, from which $(a, b) \subset (d)$ and $d = ax + by$ for some $x, y \in R$, from which $(d) \subset (a, b)$. From $r_n | r_{n-1}$ and $r_{n-2} = r_{n-1}q_n + r_n$ we find $r_n | r_{n-2}$. Continuing inductively we find $r_n | r_0, r_1$, so $r_n | b$ in E_1 . E_0 gives $r_n | a$.

Also from $r_n = r_{n-2} - r_{n-1}q_n$, $r_n \in (r_{n-1}, r_{n-2})$. Similarly, $r_{n-1} \in (r_{n-2}, r_{n-3})$, so $r_n \in (r_{n-2}, r_{n-3})$. Continuing inductively by replacing r_{k+1} in $r_n \in (r_{k+1}, r_k)$ with r_{k-1} , we find $r_n \in (a, b)$.

Let R be an integral domain not a field. Let $\tilde{R} = R - (R^\times \cup \{0\})$. An element u in $\tilde{R}$ such that for all $x \in R$, there exists r for which $x = qu + r$, where $r = 0$ or $r \in R^\times$ is called a universal side divisor. Prop: all Euclidean domains contain zero side divisors.

Let $u \in \tilde{R}$ have minimal norm. Write $x = qu + r$, where $N(r) < N(u)$ or $r = 0$. If $N(r) < N(u)$, $r \notin \tilde{R}$, so $r \in R^\times$.

P 3. Let R be a Euclidean domain and u be a nonzero element of minimal norm. Then u is a unit. Further, if $u \in R$ has $N(u) = 0$, then u is a unit.

Write $1 = qu + r$. Since $N(u)$ is minimal, $r = 0$. Hence, $qu = 1$, so u is a unit. If $N(u) = 0$, the norm of u is minimal as $N(x) \geq 0$ for all $x \in R$.

P 4. Let R be a Euclidean domain. If $(a, b) = 1$ and $a | bc$, then $a | c$. More generally, if a, b are nonzero such that $a | bc$, then $a / (a, b) | c$.

Suppose $a | b$ and $a | c$. Then $a | b + c$. Pf: write $b = ax$, $c = ay$. Then $b + c = a(x + y)$. Suppose $a | b$ and $b | c$. Then $a | c$. Write $b = ax$, $c = by$. Then $c = (ax)y = a(xy)$, or $c | a$.

Now there exist x, y such that $ax + by = 1$, so $acx + bcy = c$. $a | c - bcy$ and $a | bc$,

so $a|c$.

Write $ax + by = d$, so $acx + bcy = cd$ or $a|cd$.

P 8. Note $\mathcal{O}$ is not $\mathbb{Z}[\sqrt{D}]$. But rather: $\mathcal{O} = \mathbb{Z}[\omega]$, where $\omega = \sqrt{D}$ if $D \equiv 2, 3 \pmod{4}$ and $\omega = (1 + \sqrt{D})/2$ if $D \equiv 1 \pmod{4}$. Consider the case where $D \equiv 1 \pmod{4}$. I claim $\mathcal{O}$ is closed under multiplication. Let $\alpha, \beta \in \mathcal{O}$, so that $\alpha = \alpha_1 + \alpha_2\omega$ and $\beta = \beta_1 + \beta_2\omega$. Then $\alpha\beta = (\alpha_1 + \alpha_2\omega)(\beta_1 + \beta_2\omega) = \alpha_1\beta_1 + (\alpha_2\beta_1 + \alpha_1\beta_2)\omega + \alpha_2\beta_2\omega^2$. $\mathcal{O}$ is an additive group, so it suffices to show $\omega^2 \in \mathcal{O}$. Now

$$\left(\frac{1 + \sqrt{D}}{2}\right)^2 = \frac{1 + 2\sqrt{D} + D}{4} = \frac{D-1}{4} + \frac{1 + \sqrt{D}}{2} \in \mathcal{O},$$

so we're done.

Note that $-3, -7, -11$ are all $1 \pmod{4}$. $\mathcal{O}$ is the set of points with coordinates both integer or half-integer: if $a + b\omega \in \mathcal{O}$, if b is even, both are integer. If b is odd, both are half-integer.